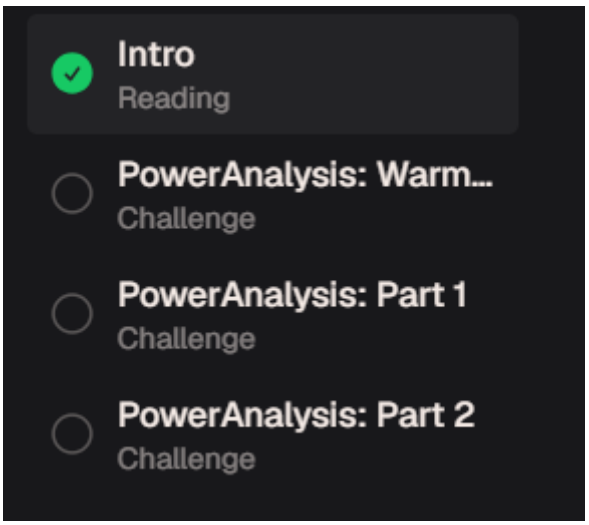
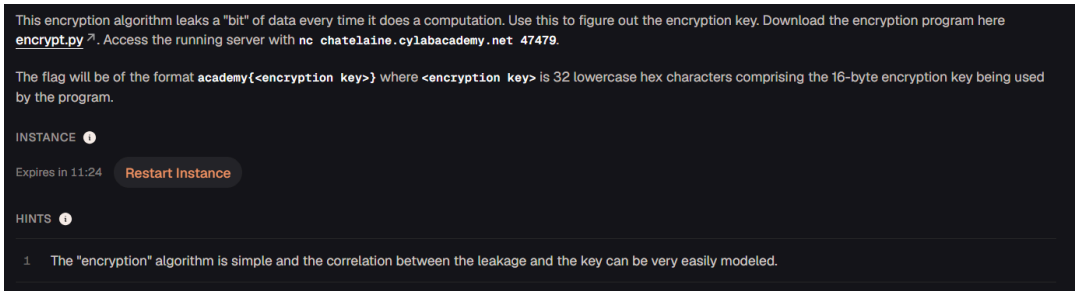


Challenge 1 : PowerAnalysis:WarmUp



File Bernama : encrypt.py

1. Kita lihat isi dari file dengan **cat encrypt.py**
2. Buat file solve bertipe python (**nano solve.py**). Source Code ada di file bernama solve.py (untuk port berubah karena waktunya habis jadi 11475)
3. Jalankan file solve tersebut (**python3 solve.py**)
4. Ketemu flagnya yakni `academy{290fda44a3d9bd690a8dba3e760f5a7e}`

```
  pos 13: kandidat akhir = ['0f']
[+] key[13] = 0f
[*] pos 14 ...
  pos 14: kandidat akhir = ['5a']
[+] key[14] = 5a
[*] pos 15 ...
  pos 15: kandidat akhir = ['7e']
[+] key[15] = 7e

[+] SECRET_KEY = 290fda44a3d9bd690a8dba3e760f5a7e
[+] Flag: academy{290fda44a3d9bd690a8dba3e760f5a7e}
```

Challenge 2 : PowerAnalysis:Part1

This embedded system allows you to measure the power consumption of the CPU while it is running an AES encryption algorithm. Use this information to leak the key via dynamic power analysis. Access the running server with **`nc chatelaine.cylabacademy.net 37746`**. It will encrypt any buffer you provide it, and output a trace of the CPU's power consumption during the operation. The flag will be of the format **`academy{<encryption key>}`** where **`<encryption key>`** is 32 lowercase hex characters comprising the 16-byte encryption key being used by the program.

INSTANCE

Expires in 14:50 Restart Instance

HINTS

1. The power consumption is correlated with the Hamming weight of the bits being processed
2. You will need to figure out how to deal with the noise

File bernama : gada hanya port (butuh waktu jir lama bat)

1. Kita coba tes lewat portnya.

```
(myenv) lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/sideChanels$ nc chatelaine.cylabacademy.net 377
46
Please provide 16 bytes of plaintext encoded as hex: 00000000000000000000000000000000
power measurement result:  [64, 82, 66, 84, 69, 67, 77, 79, 97, 101, 124, 101, 88, 83, 128, 120, 116, 69, 83, 102, 151,
160, 124, 96, 61, 60, 58, 70, 104, 109, 132, 119, 104, 96, 82, 96, 101, 74, 87, 57, 58, 40, 56, 74, 89, 114, 138, 108, 1
```

2. Buat file solve bertipe python ketika ada output plaintext dari hex yg kita masukin ada, lalu menambahkan portnya sesuai instance yang ada saat ini(**nano solve1.py**). Source Code ada di file bernama solve.py
3. Jalankan file solve tersebut (**python3 solve1.py**)

```
(myenv) lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/sideChanels$ python3 solve1.py
[*] Mengumpulkan 500 trace ...
  50/500 trace terkumpul
  100/500 trace terkumpul
  150/500 trace terkumpul
  200/500 trace terkumpul
  250/500 trace terkumpul
  300/500 trace terkumpul
  350/500 trace terkumpul
  400/500 trace terkumpul
  450/500 trace terkumpul
  500/500 trace terkumpul
[*] Shape trace: (500, 2666)
[*] CPA untuk byte 0 ...
  byte 0: key=ec, corr=0.5246, t=352
[*] CPA untuk byte 1 ...
  byte 1: key=1e, corr=0.5447, t=327
[*] CPA untuk byte 2 ...
  byte 2: key=6e, corr=0.5716, t=326
[*] CPA untuk byte 3 ...
  byte 3: key=ce, corr=0.5257, t=384
[*] CPA untuk byte 4 ...
  byte 4: key=a3, corr=0.5179, t=330
[*] CPA untuk byte 5 ...
  byte 5: key=4d, corr=0.5430, t=329
[*] CPA untuk byte 6 ...
  byte 6: key=e3, corr=0.5909, t=390
[*] CPA untuk byte 7 ...
  byte 7: key=c3, corr=0.5467, t=333
[*] CPA untuk byte 8 ...
  byte 8: key=d1, corr=0.5231, t=359
[*] CPA untuk byte 9 ...
  byte 9: key=c8, corr=0.5906, t=355
[*] CPA untuk byte 10 ...
  byte 10: key=17, corr=0.5498, t=354
[*] CPA untuk byte 11 ...
  byte 11: key=89, corr=0.5420, t=337
[*] CPA untuk byte 12 ...
  byte 12: key=bf, corr=0.5145, t=364
[*] CPA untuk byte 13 ...
  byte 13: key=49, corr=0.5561, t=339
[*] CPA untuk byte 14 ...
  byte 14: key=e4, corr=0.5357, t=358
[*] CPA untuk byte 15 ...
  byte 15: key=82, corr=0.5580, t=381

[+] SECRET_KEY = ec1e6ecea34de3c3d1c81789bf49e482
[+] Flag: academy{ec1e6ecea34de3c3d1c81789bf49e482}
```

4. Cara cepat biar ga nunggu lama gunakan **solve2.py** (ouput sama aja kek solve1.py)
5. Ketemu flagnya yakni `academy{ec1e6ecea34de3c3d1c81789bf49e482}`

Challenge 3 : PowerAnalysis:Part2

This embedded system allows you to measure the power consumption of the CPU while it is running an AES encryption algorithm. However, this time you have access to only a very limited number of measurements. Download the power-consumption traces for a sample of encryptions **traces.zip** ↗ The flag will be of the format **academy{<encryption key>}** where **<encryption key>** is 32 lowercase hex characters comprising the 16-byte encryption key being used by the program.

HINTS

1. The power consumption is correlated with the Hamming weight of the bits being processed
2. You will need to figure out how to deal with the noise

File bernama : traces.zip

1. Kita unzip untuk awalannya dengan query **unzip traces.zip**
2. Lihat file tersebut berisi apa

```
(myenv) lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/sideChanels$ cd traces
(myenv) lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/sideChanels/traces$ head -5 trace_00.txt
head: cannot open 'trace_00.txt' for reading: No such file or directory
(myenv) lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/sideChanels/traces$ head -5 trace00.txt
Plaintext: 3c99f57339dfa2a99ba7a0747e850fcc
Power trace: [138, 137, 152, 122, 96, 82, 93, 91, 89, 74, 102, 115, 115, 73, 64, 51
, 60, 74, 128, 127, 116, 93, 118, 112, 118, 98, 118, 90, 86, 51, 77, 66, 88, 93, 11
```

3. Ketika ada output panjang seperti di atas, kita langsung buat file solve sama seperti challenge 2. (**nano solve3.py**)

```
****************************************
***************THE ORACLE***************
****************************************
what should we do for you?
E --> encrypt D --> decrypt.
e
enter text to encrypt (encoded length must be less than keysize): 1
1

encoded cleartext as Hex m: 31

ciphertext (m ^ e mod n) 43746717414118196530950652036383638397057601445241916336053581346841439783210958590471265856492
728729087654320409430553992474997440703718104706823661006 89
```

4. Buat file solve bertipe python (**nano solve3.py**). Source Code ada di file bernama **solve3.py**
5. Jalankan file solve tersebut (**pythodean3 solve3.py)** dan tunggu sebentar.

```
(myenv) lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/sideChanels/traces
$ python3 solve3.py
[*] Loading traces ...
[*] Shape traces: (100, 2666)
[*] Shape plaintexts: (100, 16)
[*] CPA byte 0 ...
  byte 0: key=40, corr=0.5914, t=326
[*] CPA byte 1 ...
  byte 1: key=c4, corr=0.6093, t=327
[*] CPA byte 2 ...
  byte 2: key=57, corr=0.5393, t=360
[*] CPA byte 3 ...
  byte 3: key=de, corr=0.5791, t=327
[*] CPA byte 4 ...
  byte 4: key=8d, corr=0.5379, t=330
[*] CPA byte 5 ...
  byte 5: key=13, corr=0.6338, t=331
[*] CPA byte 6 ...
  byte 6: key=57, corr=0.5052, t=390
[*] CPA byte 7 ...
  byte 7: key=03, corr=0.5962, t=362
[*] CPA byte 8 ...
  byte 8: key=87, corr=0.6816, t=386
[*] CPA byte 9 ...
  byte 9: key=b9, corr=0.5802, t=383
[*] CPA byte 10 ...
  byte 10: key=90, corr=0.6906, t=352
[*] CPA byte 11 ...
  byte 11: key=56, corr=0.6629, t=391
[*] CPA byte 12 ...
  byte 12: key=bd, corr=0.6185, t=364
[*] CPA byte 13 ...
  byte 13: key=9a, corr=0.6554, t=359
[*] CPA byte 14 ...
  byte 14: key=17, corr=0.5783, t=382
[*] CPA byte 15 ...
  byte 15: key=d1, corr=0.5490, t=379

[+] SECRET_KEY = 40c457de8d13570387b99056bd9a17d1
[+] Flag: academy{40c457de8d13570387b99056bd9a17d1}
```

6. Ketemu flagnya yakni `academy{40c457de8d13570387b99056bd9a17d1}`